Data Communication & Computer Networks (CN)

Topic 6 of 7 : OSI Layers in Detail

Curation in progress — 122 auto-extracted questions pending verification (see Pending Review at the end)

Reference Book: **Data Communications and Networking — Behrouz A. Forouzan**

Prepared for: **RPSC Programmer • RSMSSB IA • RSMSSB SA • Data Entry/DD • BCI • SCI • REET • RAS & other Rajasthan Govt. Exams**

Total Questions: **0** verified MCQs + **122** pending review

Question Type: MCQ — every question has 4 options (A, B, C, D)

Languages: **Bilingual:** English + Hindi (हिन्दी)

Every question includes: ✔ Correct answer (why it is RIGHT) • ★ Nukta / key point (exam tip) • ✖ why each wrong option is WRONG

Verification: Every answer cross-verified **4 times** (Forouzan facts + official answer keys of Subject Test / Unit Test Solutions where available)

Syllabus Coverage — sub-topic wise order

Sub-topic curation in progress — auto-extracted questions are listed in the **"Pending Review"** section at the end.

How to read this PDF: Each question shows 4 options (A-D) in English and Hindi. The line starting with ✔ gives the correct answer and why it is right. The line starting with ★ is the **Nukta** — the key point to remember for the exam. The lines starting with ✖ explain why every wrong option is wrong. Read the question, try it yourself, then check the answer.

System: This PDF is generated by **tools/cn_build_topic.py** from **tools/data/topic6_data.py**. To add more questions, drop PDFs into **Computer Network/Incoming/** and run **tools/cn_pipeline.py** — new questions are auto-classified into topics and appended here (as pending) for review. See **Computer Network/README.md**.

Pending Review — auto-ingested, needs verification

These questions were auto-extracted & classified from uploaded PDFs. Answers/nukte to be verified before counting.

⚠ NEW — auto-ingested, answer needs verification (pending review)

P1. Which sub-layer of the Data Link Layer is responsible for flow control and error control? sub-layer flow control error control

- **(A)** MAC
- **(B)** LLC (Logical Link Control)
- **(C)** PVC
- **(D)** None of the above

[Source: BCI Subject Test-8.pdf | auto-classified: Topic 6 / Data Link: Flow & Error Control]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P2. In Go-Back-N protocol, if window size = 63, sequence number field bits required: 63

- **(A)** 5
- **(B)** 6
- **(C)** 7
- **(D)** 8

[Source: BCI Subject Test-8.pdf | auto-classified: Topic 6 / Transport Layer]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P3. A stop-and-wait ARQ channel has high latency and large bandwidth. What problem occurs? Stop-and-wait ARQ latency bandwidth

- **(A)** High throughput
- **(B)** Low channel utilization
- **(C)** CRC failure
- **(D)** Frame duplication

[Source: BCI Subject Test-8.pdf | auto-classified: Topic 6 / Data Link: Flow & Error Control]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P4. A Selective Repeat sliding window protocol uses a window size of 7 with 3-bit sequence numbers. Is it valid? Sliding window sequence number 3-bit window size 7 valid

- **(A)** Valid
- **(B)** Invalid – violates maximum window rule
- **(C)** Valid only for piggybacking
- **(D)** Valid only with selective repeat

[Source: BCI Subject Test-8.pdf | auto-classified: Topic 6 / Data Link: Flow & Error Control]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P5. In a noisy channel, Hamming code is used with a minimum distance of 3. What can it guarantee? Noisy Hamming code minimum distance 3 guarantee

(A) Detect 2-bit errors
(B) Correct 2-bit errors
(C) Correct 3-bit errors
(D) Detect 4-bit errors

[Source: BCI Subject Test-8.pdf | auto-classified: Topic 6 / Data Link: Flow & Error Control]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P6. A Go-Back-N sender uses window size 4. Frame 2 is lost. What happens after receiver receives frames 3 and 4? Go-Back- N window size 4 Frame 2 Frame 3 4 receiver

(A) Stores them in buffer
(B) Sends ACK for 4
(C) Discards frames 3 and 4
(D) Acts like selective repeat

[Source: BCI Subject Test-8.pdf | auto-classified: Topic 6 / Data Link: Flow & Error Control]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P7. A client opens a TCP connection to a web server on port 80. The client uses port 49152. What is the complete socket pair? TCP connection port 80 port 49152 socket pair

(A) 192.168.1.10:49152 → 172.16.0.5:80
(B) 192.168.1.10:80 → 172.16.0.5:49152
(C) 49152 → 80
(D) 80 → 49152

[Source: BCI Subject Test-8.pdf | auto-classified: Topic 6 / Transport Layer]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P8. A server receives a packet with destination port 443. This implies: destination port 443 packet

(A) HTTP request
(B) HTTPS traffic
(C) FTP command
(D) Email traffic

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⚠ NEW — auto-ingested, answer needs verification (pending review)

P9. Which port is used by DHCP client when requesting an IP address? DHCP client IP request - port

(A) UDP 67
(B) UDP 68
(C) TCP 68
(D) TCP 67

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⚠ NEW — auto-ingested, answer needs verification (pending review)

P10. Which protocol must use both TCP and UDP for different operations? TCP UDP

(A) DNS
(B) HTTP
(C) POP3
(D) Telnet

[Source: BCI Subject Test-8.pdf | auto-classified: Topic 6 / Transport Layer]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P11. A firewall blocks all ports except 22. What service remains accessible? Firewall ports block 22

(A) FTP
(B) SSH
(C) HTTPS
(D) SMTP

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⚠ NEW — auto-ingested, answer needs verification (pending review)

P12. Which of the following technique is used to verify the integrity of the message? # % B

(A) Message Encrypt
(B) Message Checksum
(C) Message Digest
(D) None of these

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 6 / Data Link: Flow & Error Control]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P13. Which of the following functions are performed by the Data Link Layer? ‰ k l B - P u? 1.Framing / Š 2. Error Detection / ‹ 3. Flow Control / Œ { 4. Path Determination / P

(A) 1, 2 3
(B) 1 4
(C) 2 4
(D) 1, 2, 3 4

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 6 / Data Link: Flow & Error Control]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P14. Which of the following statements about MAC and LLC sublayers is TRUE? MAC LLC - B ƒ ? 1. MAC sublayer handles physical addressing. MAC - { 2. LLC sublayer provides error control and flow control. LLC - ‹ Œ { @ 3. MAC sublayer performs routing. MAC - 2

(A) 1 2
(B) 2 3
(C) 1 3
(D) 1, 2 3

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 6 / Data Link: Flow & Error Control]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P15. In Ethernet, the minimum frame size is fixed mainly to: B Ž Š v 2 •

(A) Improve throughput
(B) Reduce overhead
(C) Detect collisions using CSMA/CD
(D) Increase security

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 6 / Data Link: Framing & MAC]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P16. Collision detection and avoidance mechanisms are implemented in: (Collision) ‰ B

(A) LLC Sublayer
(B) Network Layer
(C) MAC Sublayer
(D) Physical Layer

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 6 / Data Link: Framing & MAC]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P17. Which of the following is NOT a responsibility of the LLC sublayer? # B LLC - 9

(A) Flow control
(B) Error control
(C) Service interface to Network layer
(D) Physical addressing

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 6 / Data Link: Flow & Error Control]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P18. Which protocol requires time synchronization among nodes

(A) Pure ALOHA
(B) Slotted ALOHA
(C) CSMA
(D) CSMA/CD

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 6 / Data Link: Framing & MAC]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P19. Which protocol detects collisions while transmission is in progress? collision detect

(A) CSMA
(B) Slotted ALOHA
(C) CSMA/CD
(D) CSMA/CA

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 6 / Data Link: Framing & MAC]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P20. Why is CSMA/CD not suitable for wireless networks? 9 B CSMA/CD – —

(A) No random access
(B) Collision detection is not feasible in wireless medium
(C) High bandwidth requirement
(D) Requires slot synchronization

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 6 / Data Link: Framing & MAC]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P21. Which mechanism is used by CSMA/CA to reduce collisions? CSMA/CA B collision X

(A) Collision detection
(B) Slot restriction
(C) RTS/CTS handshake
(D) Jam signal

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 6 / Data Link: Framing & MAC]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P22. The vulnerable period of Pure ALOHA is: / Pure ALOHA vulnerable period

(A) T
(B) 2T
(C) T/2
(D) 4T

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⚠ NEW — auto-ingested, answer needs verification (pending review)

P23. Which protocol has the highest efficiency under heavy load? %‘ B

(A) Pure ALOHA
(B) Slotted ALOHA
(C) CSMA
(D) CSMA/CD

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⚠ NEW — auto-ingested, answer needs verification (pending review)

P24. Why is bit stuffing required in bit- oriented protocols? Bit-oriented - B bit stuffing — ¢

(A) To increase bandwidth
(B) To prevent confusion between data and flag pattern
(C) To reduce collisions
(D) To detect errors

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 6 / Data Link: Framing & MAC]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P25. The minimum size of an Ethernet frame (excluding preamble) is: £ Ethernet ¦ §

(A) 46 bytes
(B) 64 bytes
(C) 72 bytes
(D) 1518 bytes

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 6 / Data Link: Framing & MAC]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P26. The correct order of fields in an Ethernet frame is: Ethernet ¦ B ¨ ‰ —

(A) Destination → Source → FCS → Data→Preamble
(B) Preamble → Destination → Source → Type/Length → Data → CRC
(C) Source → Destination → Data → CRC→Type →Preamble
(D) Preamble → Source → Destination → Data

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 6 / Data Link: Framing & MAC]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P27. The data field size in an Ethernet frame ranges from: Ethernet ¦ B ¨ B

(A) 64–1518 bytes
(B) 46–1500 bytes
(C) 40–1480 bytes
(D) 32–1500 bytes

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 6 / Data Link: Framing & MAC]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P28. The Frame Check Sequence (FCS) field is used for: FCS (Frame Check Sequence) ¨

(A) Flow control
(B) Address resolution
(C) Error detection using CRC
(D) Medium access

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 6 / Data Link: Flow & Error Control]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P29. The minimum Ethernet frame size is necessary to support: Ethernet § ¦ 9 ¢

(A) Flow control
(B) Error correction
(C) CSMA/CD collision detection
(D) Routing

[Source: BCI Subject Test-9.pdf | auto-classified: Topic 6 / Data Link: Framing & MAC]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P30. Which of the following is an example of Client–Server architecture? H I + Client–Server 9 {

(A) Chat between two Bluetooth devices
(B) Web browser accessing a website
(C) File transfer via USB cable
(D) Two laptops connected directly

[Source: BCI_SCI Subject Test-01 (CN and Security).pdf | auto-classified: Topic 6 / Session, Presentation & Application]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P31. Which of the following is NOT a function of Data Link layer? A ?**

(A) Framing
(B) Error detection
(C) Error correction
(D) Routing

[Source: BCI_SCI Subject Test-7.pdf | auto-classified: Topic 6 / Data Link: Flow & Error Control]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P32. Which protocol is used for path MTU discovery? MTU b W ?**

(A) TCP
(B) UDP
(C) ICMP
(D) ARP

[Source: BCI_SCI Subject Test-7.pdf | auto-classified: Topic 6 / Network Layer]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P33. Which of the following is a connectionless transport service? 9 7 t E A ?**

(A) TCP
(B) ARP
(C) UDP
(D) FTP

[Source: BCI_SCI Subject Test-7.pdf | auto-classified: Topic 6 / Transport Layer]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P34. How many bits are used for port addressing? A E O?**

(A) 8
(B) 10
(C) 16
(D) 32

[Source: BCI_SCI Subject Test-7.pdf | auto-classified: Topic 6 / Transport Layer]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P35. Which protocol uses different ports for control and data? W Control Data A ?**

(A) FTP
(B) HTTP
(C) DNS
(D) Telnet

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⚠ NEW — auto-ingested, answer needs verification (pending review)

P36. Which tool is used to view MAC address in a system? n 7 MAC E ?**

(A) ping
(B) ipconfig /all
(C) netstat
(D) arp -a

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⚠ NEW — auto-ingested, answer needs verification (pending review)

P37. What is the function of ARP?ARP

(A) Resolve IP to MAC address
(B) Resolve domain name
(C) Encrypt packets
(D) Determine route

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⚠ NEW — auto-ingested, answer needs verification (pending review)

P38. Which field in IP header helps in fragmentation?IP

(A) TTL
(B) Header length
(C) Identification
(D) Protocol

[Source: CN Topic Test 6.pdf | auto-classified: Topic 6 / Network Layer]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P39. Which of these protocols supports f ile transfer

(A) FTP
(B) HTTP © DNS (d) ARP 10. The term MTU stands for?MTU (a). Maximum Transmit Unit (b) Maximum Transfer Utility
(C) Maximum Transmission Unit
(D) Minimum Transmission Unit

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⚠ NEW — auto-ingested, answer needs verification (pending review)

P40. Which of these protocols work at data link layer

(A) IP
(B) TCP
(C) Ethernet
(D) UDP

[Source: CN Topic Test 6.pdf | auto-classified: Topic 6 / Transport Layer]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P41. Which of the following uses port number 25? 25

(A) FTP
(B) HTTP
(C) SMTP
(D) Telnet

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⚠ NEW — auto-ingested, answer needs verification (pending review)

P42. Which field in the TCP header is used for flow control?TCP

(A) Window Size
(B) Sequence Number
(C) Acknowledgment Number
(D) Checksum

[Source: CN Topic Test 6.pdf | auto-classified: Topic 6 / Transport Layer]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P43. How does RIP prevent routing loops?RIP

(A) Poison reverse
(B) Sequence numbers
(C) Split horizon
(D) Both A and C

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⚠ NEW — auto-ingested, answer needs verification (pending review)

P44. What is the purpose of the SYN flag in TCP?TCP SYN

(A) To end a session
(B) To initiate a connection
(C) To reset connection
(D) For data transmission

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⚠ NEW — auto-ingested, answer needs verification (pending review)

P45. Which one is a distance-vector routing protocol

(A) RIP
(B) OSPF
(C) BGP
(D) IS-IS

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⚠ NEW — auto-ingested, answer needs verification (pending review)

P46. Which of these protocols allows loop-free switching in Layer 2 networks? 2

(A) RIP
(B) OSPF
(C) STP(SPANNING TREE PROTOCOL)
(D) EIGRP

[Source: CN Topic Test 6.pdf | auto-classified: Topic 6 / Network Layer]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P47. In TCP, which state follows the SYN- SENT state?TCP SYN-SENT

(A) SYN-RECEIVED
(B) LISTEN
(C) ESTABLISHED
(D) CLOSED

[Source: CN Topic Test 6.pdf | auto-classified: Topic 6 / Transport Layer]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P48. What is a socket in networking

(A) IP + MAC
(B) IP + Port
(C) MAC + Port
(D) IP + DNS

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⚠ NEW — auto-ingested, answer needs verification (pending review)

P49. What protocol is used to find the MAC address of a given IP? IP MAC

(A) DHCP
(B) ARP
(C) RARP
(D) SNMP

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⚠ NEW — auto-ingested, answer needs verification (pending review)

P50. TCP uses which type of congestion control technique initially?TCP

(A) Congestion avoidance
(B) Fast recovery
(C) Slow start
(D) Leaky bucket

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⚠ NEW — auto-ingested, answer needs verification (pending review)

P51. Which algorithm is used by OSPF to compute the shortest path?OSPF

(A) Bellman-Ford
(B) Dijkstra
(C) A*
(D) Kruskal

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⚠ NEW — auto-ingested, answer needs verification (pending review)

P52. How many maximum hops are allowed in RIP?RIP

(A) 8
(B) 16
(C) 15
(D) 32

[Source: CN Topic Test 6.pdf | auto-classified: Topic 6 / Network Layer]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P53. What is fragmentation in IP networking?IP

(A) Splitting data into packets
(B) Splitting large packets for transmission
(C) Combining multiple packets
(D) Encrypting packets

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⚠ NEW — auto-ingested, answer needs verification (pending review)

P54. Which port does Telnet use by default?Telnet

(A) 21
(B) 23
(C) 22
(D) 25

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⚠ NEW — auto-ingested, answer needs verification (pending review)

P55. Which of the following protocols is not secure

(A) HTTPS
(B) FTP
(C) SSH
(D) SFTP

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⚠ NEW — auto-ingested, answer needs verification (pending review)

P56. Which protocol is used for secure shell login

(A) FTP
(B) SSH
(C) Telnet
(D) SNMP

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⚠ NEW — auto-ingested, answer needs verification (pending review)

P57. What kind of delay does TCP handle using its window size?TCP

(A) Transmission delay
(B) Queuing delay
(C) Propagation delay
(D) Bandwidth delay product

[Source: CN Topic Test 6.pdf | auto-classified: Topic 6 / Transport Layer]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P58. What does a "ping" command test? "ping"

(A) IP Configuration
(B) Host reachability
(C) DNS resolution
(D) Port scanning 44 Which application layer protocol is used for email sending? (a) POP3 (b) SMTP (c) IMAP (d) SNMP

[Source: CN Topic Test 6.pdf | auto-classified: Topic 6 / Session, Presentation & Application]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P59. Which technique uses both acknowledgment and sequence numbers

(A) Go-Back-N
(B) Stop and Wait
(C) Selective Repea
(D) all of the above

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⚠ NEW — auto-ingested, answer needs verification (pending review)

P60. In which protocol is "port knocking" a security mechanism?" "

(A) TCP
(B) UDP
(C) SSH
(D) IP

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⚠ NEW — auto-ingested, answer needs verification (pending review)

P61. Which of the following supports QoS? - QoS (Quality of Service)

(A) TCP
(B) UDP
(C) MPLS
(D) ARP

[Source: CN Topic Test 6.pdf | auto-classified: Topic 6 / Transport Layer]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P62. What is the main purpose of the Sliding Window in TCP ? TCP

(A) Address resolution

(B) Flow control

(C) Error detection

(D) Encryption

[Source: CN Topic Test 5.pdf | auto-classified: Topic 6 / Data Link: Flow & Error Control]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P63. Which of the following is not a valid private IP address? IP

(A) 10.0.0.1

(B) 172.16.5.4 (a) 192.168.1.10 (d)11.0.0.1 3. Which of the following is used for congestion control in TCP? TCP (a) Slow start (a) Leaky bucket (b) Token bucket (d) Round Robin 4. What is the maximum size of a TCP header without options?TCP ( ) (a) 16 bytes (b) 20 bytes

(C) 24 bytes

(D) 32 bytes

[Source: CN Topic Test 5.pdf | auto-classified: Topic 6 / Transport Layer]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P64. BGP uses which algorithm to make routing decisions?BGP

(A) Bellman-Ford

(B) Dijkstra

(C) Mailing list

(D) Hypertext markup language

[Source: CN Topic Test 5.pdf | auto-classified: Topic 6 / Network Layer]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P65. What is meant by "persistent HTTP connection"?"Persistent HTTP "

(A) Connection is always secure / Sorting data on a disk drive (a) Sending information to a host computer

(B) Receiving information from a host computer (d) Sorting data on a hard drive 8. What is the minimum time TCP must wait before retransmitting a packet?TCP (a) RTT/2 (b) Timeout value

(C) 1 Second

(D) ACK Timeout

[Source: CN Topic Test 5.pdf | auto-classified: Topic 6 / Transport Layer]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P66. Which of the following protocols does not operate at the application layer

(A) DNS
(B) FTP
(C) ARP
(D) SMTP

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⚠ NEW — auto-ingested, answer needs verification (pending review)

P67. Which of the following is NOT a valid ICMP error message? ICMP

(A) Destination Unreachable
(B) Time Exceeded
(C) Echo Request
(D) Port Blocked

[Source: CN Topic Test 5.pdf | auto-classified: Topic 6 / Network Layer]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P68. What is the primary role of CSMA/CD in Ethernet?Ethernet CSMA/CD

(A) Error Correction
(B) Collision Detection
(C) Address Resolution
(D) Flow Control

[Source: CN Topic Test 5.pdf | auto-classified: Topic 6 / Data Link: Framing & MAC]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P69. In which scenario will TCP's three- way handshake fail?TCP

(A) Firewall blocks SYN-ACK
(B) High latency
(C) ACK lost
(D) Server is busy

[Source: CN Topic Test 5.pdf | auto-classified: Topic 6 / Transport Layer]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P70. What is the purpose of TCP congestion window (cwnd)?TCP (cwnd)

(A) Tracks unacknowledged packets
(B) Controls flow rate
(C) Handles encryption
(D) Prioritize packets

[Source: CN Topic Test 5.pdf | auto-classified: Topic 6 / Transport Layer]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P71. Which one of the following attacks exploits TCP's 3-way handshake? TCP

(A) SYN Flood
(B) DNS Spoofing
(C) Man in the Middle
(D) ARP Poisoning

[Source: CN Topic Test 5.pdf | auto-classified: Topic 6 / Transport Layer]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P72. What does a “traceroute” command do?“traceroute”

(A) Lists all devices on a network
(B) Tracks the path a packet takes
(C) Resolves domain names
(D) Opens ports on firewall

[Source: CN Topic Test 5.pdf | auto-classified: Topic 6 / Network Layer]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P73. Which field in TCP header ensures data integrity?TCP

(A) Urgent Pointer
(B) Sequence Number
(C) Checksum
(D) Acknowledgment Number

[Source: CN Topic Test 5.pdf | auto-classified: Topic 6 / Transport Layer]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P74. Which of the following is not a property of TCP? TCP

(A) Reliable
(B) Ordered
(C) Connection-oriented
(D) Broadcast

[Source: CN Topic Test 5.pdf | auto-classified: Topic 6 / Transport Layer]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P75. Which protocol does traceroute use on UNIX/Linux systems?UNIX/Linux traceroute

(A) TCP
(B) UDP
(C) IMCP Echo
(D) IGMP

[Source: CN Topic Test 5.pdf | auto-classified: Topic 6 / Network Layer]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P76. What type of attack is prevented by using random initial sequence numbers in TCP?TCP

(A) Smurf attack
(B) SYN Flooding
(C) TCP Session hijacking
(D) Ping of Death

[Source: CN Topic Test 5.pdf | auto-classified: Topic 6 / Transport Layer]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P77. Which command is used to see routing table in Linux?Linux

(A) ifconfig
(B) netstat- r
(C) Ping
(D) ipconfig Which protocol is connection- oriented? (a) UDP (b) IP (c) TCP (d) ARP

[Source: CN Topic Test 5.pdf | auto-classified: Topic 6 / Network Layer]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P78. What is the MTU size for Ethernet?Ethernet MTU

(A) 1024 bytes
(B) 1500 bytes
(C) 512 bytes
(D) 2048

[Source: CN Topic Test 5.pdf | auto-classified: Topic 6 / Data Link: Framing & MAC]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P79. Which header field in TCP ensures reliable order?TCP

(A) Checksum
(B) Flags
(C) Sequence Number
(D) Port Number

[Source: CN Topic Test 5.pdf | auto-classified: Topic 6 / Transport Layer]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P80. Which of the following uses a logical addressing scheme

(A) MAC
(B) IP
(C) ARP
(D) DNS Which protocol provides reliable delivery using sequence numbers? (a) UDP (b) IP (c) TCP (d) ARP

[Source: CN Topic Test 5.pdf | auto-classified: Topic 6 / Network Layer]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P81. Which of these is a connectionless protocol

(A) TCP
(B) FTP
(C) UDP
(D) HTTP

[Source: CN Topic Test 5.pdf | auto-classified: Topic 6 / Transport Layer]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P82. Which protocol translates domain names to IP addresses? IP

(A) DHCP
(B) DNS
(C) HTTP
(D) FTP

[Source: CN Topic Test 5.pdf | auto-classified: Topic 6 / Session, Presentation & Application]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P83. How many digits of the Data Network Identification Code (DNIC) identify the country? (DNIC)

(A) first three
(B) first four
(C) first five
(D) first six

[Source: CN Topic Test-01.pdf | auto-classified: Topic 6 / Network Layer]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P84. Which of the following methods is used to broadcast two packets on the medium at a time

(A) Collision
(B) Synchronous
(C) Asynchronous
(D) None of the above

[Source: CN Topic Test-01.pdf | auto-classified: Topic 6 / Data Link: Framing & MAC]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P85. Which protocol is primarily used for secure web browsing

(A) FTP
(B) HTTP
(C) HTTPS
(D) SMTP

[Source: CN Topic Test-01.pdf | auto-classified: Topic 6 / Session, Presentation & Application]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P86. Which one of the following is not used in media access control

(A) ethernet
(B) digital subscriber line
(C) fiber distributed data interface
(D) none of the mentioned

[Source: CN Topic Test-01.pdf | auto-classified: Topic 6 / Data Link: Framing & MAC]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P87. Application layer is implemented in

(A) End system
(B) NIC
(C) Ethernet
(D) None of the mentioned

[Source: CN Topic Test-02.pdf | auto-classified: Topic 6 / Data Link: Framing & MAC]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P88. The functionalities of presentation layer includes

(A) Data compression
(B) Data encryption
(C) Data description
(D) All of the mentioned

[Source: CN Topic Test-02.pdf | auto-classified: Topic 6 / Session, Presentation & Application]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P89. Which protocol layer uses the protocols are WWW, HTTP, FTP, SMTP, email, etc ? WWW, HTTP, FTP, SMTP

(A) Application Layer
(B) Transport Layer
(C) Internet Layer
(D) Hardware Layer

[Source: CN Topic Test-02.pdf | auto-classified: Topic 6 / Session, Presentation & Application]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P90. Which protocol is generally used to encrypt data between web browser and web server

(A) SSL
(B) Digital Signature
(C) IPSec
(D) TCP

[Source: CN Topic Test-02.pdf | auto-classified: Topic 6 / Session, Presentation & Application]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P91. Which of the following is NOT a responsibility of the network layer

(A) Congestion control
(B) Error control
(C) Routing
(D) Inter-networking

[Source: CN Topic Test-02.pdf | auto-classified: Topic 6 / Data Link: Flow & Error Control]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P92. What term is used for an endpoint in an inter-process communication across a network

(A) Port
(B) Machine
(C) Socket
(D) Pipe

[Source: CN Topic Test-02.pdf | auto-classified: Topic 6 / Transport Layer]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P93. _____ is the process of detecting and correcting the error. While _____ is a mechanism to ensure the efficient delivery of Dat

(A) _____ _____ (a) Error control, Flow control
(B) Process control, Network control
(C) Session control, Flow control
(D) OSI Control, Data control

[Source: CN Topic Test-02.pdf | auto-classified: Topic 6 / Data Link: Flow & Error Control]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P94. In one of the pairs of protocols given below, both the protocols can use multiple TCP connections between the same client and the server. Which one is that

(A) HTTP, FTP
(B) HTTP, TELNET
(C) FTP, SMTP
(D) HTTP, SMTP

[Source: CN Topic Test-02.pdf | auto-classified: Topic 6 / Session, Presentation & Application]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P95. Transport layer protocol is used to support electronic mail

(A) SNMP
(B) IP
(C) SMTP
(D) TCP

[Source: CN Topic Test-03.pdf | auto-classified: Topic 6 / Session, Presentation & Application]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P96. Identify the true statements from the given statements. (1)HTTP may use different TCP connection for different objects of a webpage if non persistent connections are used (2) FTP uses two TCP connections, one for data and another control (3)TELNET and FTP can only use TWO connection at a time. (1) HTTP TCP (2) FTP TCP (3) TELNET FTP

(A) (1)
(B) (1) and (2)
(C) (2) and (3)
(D) (1),(2) and (3)

[Source: CN Topic Test-03.pdf | auto-classified: Topic 6 / Session, Presentation & Application]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P97. Which protocol is used for dynamic IP address assignment

(A) DNS
(B) ARP
(C) DHCP
(D) FTP

[Source: CN Topic Test-03.pdf | auto-classified: Topic 6 / Network Layer]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P98. What is the value of acknowledgement field in segment

(A) Number of previous bytes to receive
(B) Total number of bytes to receive
(C) Number of next bytes to receive
(D) Sequence of zero's and one's

[Source: CN Topic Test-04.pdf | auto-classified: Topic 6 / Data Link: Flow & Error Control]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P99. In Stop-and-Wait ARQ, after sending one frame, the sender: ARQ

(A) Waits for acknowledgment
(B) Sends the next frame immediately
(C) Stops transmission permanently
(D) Sends a NAK

[Source: CN Topic Test-04.pdf | auto-classified: Topic 6 / Data Link: Flow & Error Control]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P100. Which protocol allows the sender to transmit multiple frames before needing an acknowledgment

(A) Stop-and-Wait
(B) Go-Back-N
(C) ALOHA
(D) CSMA/CD

[Source: CN Topic Test-04.pdf | auto-classified: Topic 6 / Data Link: Framing & MAC]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P101. What is the window size in Stop- and-Wait ARQ

(A) 0
(B) 1
(C) N
(D) 2

[Source: CN Topic Test-04.pdf | auto-classified: Topic 6 / Transport Layer]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P102. In Go-Back-N ARQ, if an error is detected in a frame: Go-Back-N ARQ , :

(A) Only that frame is retransmitted
(B) All frames are ignored
(C) All frames from that one onward are retransmitted
(D) No frame is retransmitted

[Source: CN Topic Test-04.pdf | auto-classified: Topic 6 / Data Link: Framing & MAC]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P103. What is the main disadvantage of Stop-and-Wait flow control

(A) Complexity
(B) Low efficiency due to waiting
(C) High memory usage
(D) Frame loss

[Source: CN Topic Test-04.pdf | auto-classified: Topic 6 / Data Link: Flow & Error Control]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P104. The maximum number of outstanding frames in Selective Repeat is: Outstanding

(A) N
(B) N – 1
(C) N/2
(D) 2N

[Source: CN Topic Test-04.pdf | auto-classified: Topic 6 / Data Link: Flow & Error Control]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P105. Which protocol has better bandwidth utilization

- **(A)** Stop-and-Wait
- **(B)** Selective Repeat
- **(C)** Go-Back-N
- **(D)** None of the above

[Source: CN Topic Test-04.pdf | auto-classified: Topic 6 / Data Link: Flow & Error Control]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P106. Flow control prevents which of the following issues

- **(A)** Deadlock
- **(B)** Congestion collapse
- **(C)** Overflow of the receiver's buffer
- **(D)** Packet routing error

[Source: CN Topic Test-04.pdf | auto-classified: Topic 6 / Transport Layer]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P107. Which of the following is not a function of the Data Link Layer

- **(A)** Framing
- **(B)** Error control
- **(C)** Routing
- **(D)** Flow control

[Source: CN Topic Test-04.pdf | auto-classified: Topic 6 / Data Link: Flow & Error Control]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P108. Which error detection method uses one extra bit added to the end of a byte

- **(A)** CRC
- **(B)** Checksum
- **(C)** Hamming code
- **(D)** Parity bit

[Source: CN Topic Test-04.pdf | auto-classified: Topic 6 / Data Link: Flow & Error Control]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P109. In Stop-and-Wait ARQ, what happens when a frame is lost? ARQ ,

- **(A)** Receiver sends NAK
- **(B)** Sender retransmits after timeout
- **(C)** Receiver retransmits
- **(D)** No retransmission

[Source: CN Topic Test-04.pdf | auto-classified: Topic 6 / Data Link: Flow & Error Control]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P110. Virtual circuit is associated with a

(A) Connectionless

(B) Error-free

(C) Segmentation

(D) Connection-oriented service.

[Source: CN Topic Test-04.pdf | auto-classified: Topic 6 / Transport Layer]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P111. The cost of the network is usually determined by: :

(A) Time complexity

(B) Switching complexity

(C) Circuit complexity

(D) None of these Which of these handles reliable data transfer over a link(Node to Node) ? ((a) Network Layer (b) TCP (c) ARQ (d) FTP

[Source: CN Topic Test-04.pdf | auto-classified: Topic 6 / Transport Layer]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P112. Which of the following is an error detection technique

(A) ARQ

(B) Parity Bit

(C) Stop-and-Wait

(D) Sliding Window 40 Which of the following is responsible for framing? (a) Transport Layer (b) Network Layer (c) Physical Layer (d) Data Link Layer Which of the following is an Ethernet connector type? (a) RJ-45 (b) RJ-69 (c) RJ-11 (d) RJ-15 42 A firewall in a Computer Network is a (a) system to prevent unauthorized access. (b) Physical boundary of network (C) simple web browsing (d) None of these The _____ layer is responsible for the delivery of a message from one process to another. _____ (a)physical (b)transport (c)network (d) application 44 What is the maximum supported distance for an Ethernet cable run? (a) 150 m / 492 ft (b) 50m / 164 ft (c) 100m / 328 ft (d) 200 m / 656 ft Which of the following is NOT an application layer protocol for email? (a) TCP (b) IMAP (c) POP (d) SMTP 46 The process in which information packets are forwarded to hops is called: (a) Communication (b) Switching (c) Routing (d) None of these Which switching type has the re- route capability: (a) Circuit Switching (b) Packet Switching (c) Switching (d) None of them ____ protocol is used by Telnet. ____ (a) IP (b) TCP (c) UDP (d) DHCP 49 A server that has HTTP services and SMTP services bound to an interface functions with what protocol? HTTP SMTP (a) DHCP (b) UDP (c) TCP (d) IP What is the subnet mask of this network: 192.168.0.0/29

[Source: CN Topic Test-04.pdf | auto-classified: Topic 6 / Session, Presentation & Application]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P113. For Stop-and-Wait ARQ, for 10 data packets sent, ... acknowledgements are needed.

(A) 9
(B) 20
(C) 11
(D) 10

[Source: RPSC Programmer 2024 Paper-1 (with official key) | auto-classified: Topic 6 / Data Link: Flow & Error Control]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P114. The message 1001 is to be transmitted using the CRC polynomial $x^3 + x + 1$ to protect it from errors. The message that should be transmitted is -

(A) 1001100
(B) 1001011
(C) 1001110
(D) 1001000

[Source: RPSC Programmer 2024 Paper-1 (with official key) | auto-classified: Topic 6 / Data Link: Flow & Error Control]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P115. In a slotted ALOHA with G-stations attempting to transmit in one slot, the throughput is equal to -

(A) $G \times e^{-G}$
(B) $2G \times e^{-G}$
(C) e^{-G}
(D) $G \times e^{-2G}$

[Source: RPSC Programmer 2024 Paper-1 (with official key) | auto-classified: Topic 6 / Data Link: Framing & MAC]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P116. Which of the following protocol is used for transferring email messages from one machine to another?

(A) SNMP
(B) FTP
(C) TELNET
(D) SMTP

[Source: RPSC Programmer 2024 Paper-1 (with official key) | auto-classified: Topic 6 / Session, Presentation & Application]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P117. Which of the following Multiple Access Protocols is not suitable for wired networks?

(A) CSMA/CD
(B) TGMA
(C) TDMA
(D) CSMA/CA

[Source: RPSC Programmer 2024 Paper-1 (with official key) | auto-classified: Topic 6 / Data Link: Framing & MAC]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P118. The three-way handshake mechanism in TCP aims to -

(A) secure the communication
(B) send bits from one device to another over an optical fibre connection
(C) perform error detection and correction
(D) establish connection and synchronize sequence numbers

[Source: RPSC Programmer 2024 Paper-1 (with official key) | auto-classified: Topic 6 / Transport Layer]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P119. Which of the following protocol may be used as zeroconf and plug-and-play?

(A) TCP
(B) DHCP
(C) NAT
(D) ICMP

[Source: RPSC Programmer 2024 Paper-1 (with official key) | auto-classified: Topic 6 / Network Layer]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P120. How many bits for 'Fragment Offset' field are there in IPv4 header?

(A) 16 bits
(B) 10 bits
(C) 13 bits
(D) 32 bits

[Source: RPSC Programmer 2024 Paper-1 (with official key) | auto-classified: Topic 6 / Network Layer]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P121. Network Allocation Vector (NAV) is associated with which of the following MAC technologies?

(A) Wi-Fi
(B) CSMA/CD
(C) Bluetooth
(D) Ethernet

[Source: RPSC Programmer 2024 Paper-1 (with official key) | auto-classified: Topic 6 / Data Link: Framing & MAC]

⚠ NEW — auto-ingested, answer needs verification (pending review)

P122. Which of the following is not related to data link layer?

(A) Error control

(B) Congestion control

(C) Framing

(D) MAC addressing

[Source: RPSC Programmer 2024 Paper-1 (with official key) | auto-classified: Topic 6 / Data Link: Flow & Error Control]